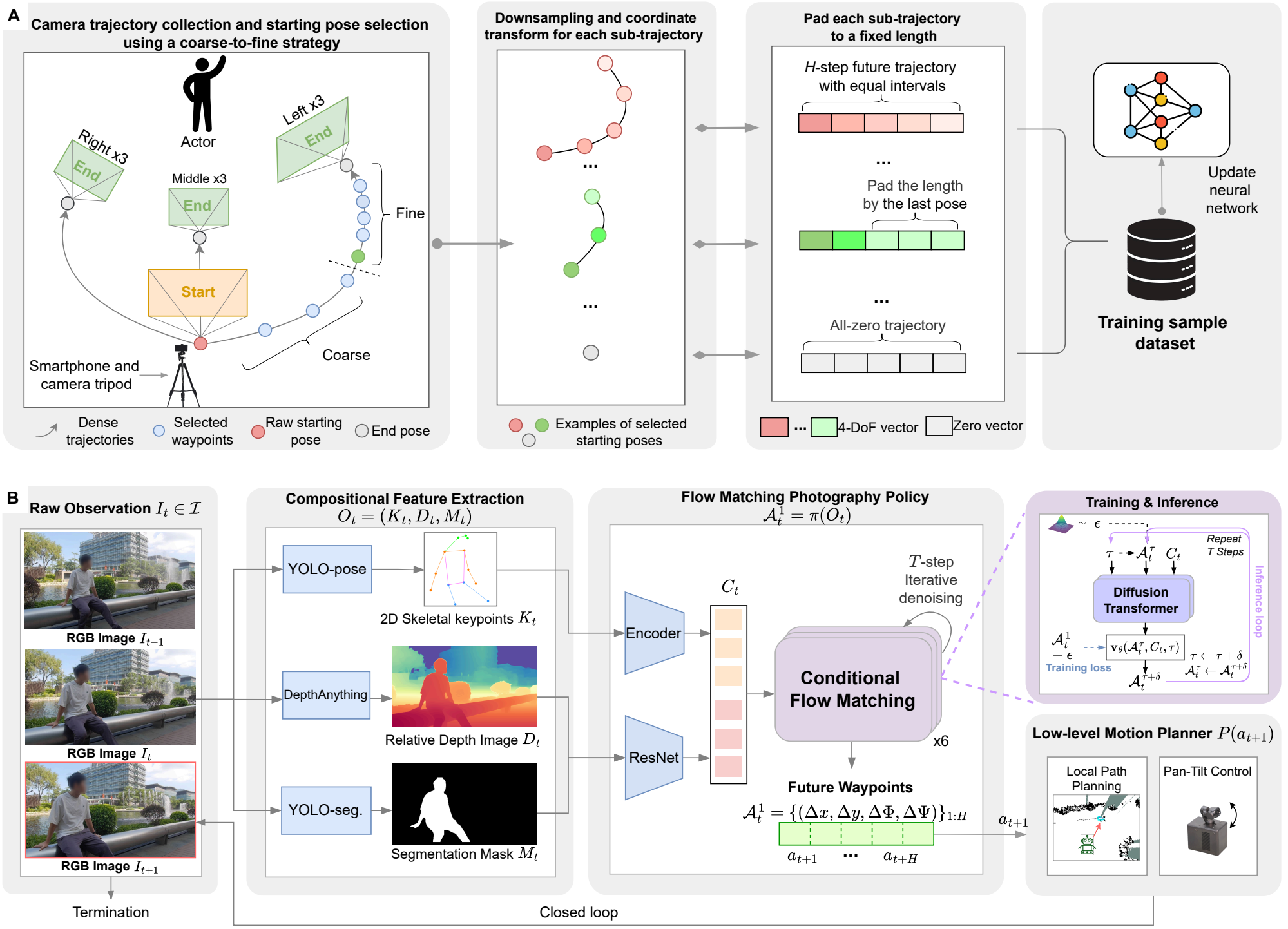
A
Camera trajectory collection and starting pose selection using a coarse-to-fine strategy
Actor
Right x3
Left x3
Middle x3
End
Start
Fine
Coarse
Smartphone and camera tripod
Dense trajectories
Selected waypoints
Raw starting pose
End pose
Downsampling and coordinate transform for each sub-trajectory
Examples of selected starting poses
Pad each sub-trajectory to a fixed length
H-step future trajectory with equal intervals
Pad the length by the last pose
All-zero trajectory
4-DoF vector
Zero vector
Update neural network
Training sample dataset
B
Raw Observation $I_t \in \mathcal{I}$
RGB Image I_{t-1}
RGB Image I_t
RGB Image I_{t+1}
Termination
Compositional Feature Extraction
$O_t = (K_t, D_t, M_t)$
YOLO-pose
2D Skeletal keypoints K_t
DepthAnything
Relative Depth Image D_t
YOLO-seg.
Segmentation Mask M_t
Flow Matching Photography Policy
$\mathcal{A}_t^1 = \pi(O_t)$
Encoder
ResNet
C_t
T-step Iterative denoising
Conditional Flow Matching
x6
Future Waypoints
$\mathcal{A}_t^1 = \{(\Delta x, \Delta y, \Delta\Phi, \Delta\Psi)\}_{1:H}$
a_{t+1} ... a_{t+H}
a_{t+1}
Training & Inference
Repeat T Steps
Diffusion Transformer
Inference loop
Training loss
$\mathbf{v}_\theta(\mathcal{A}_t^\tau, C_t, \tau)$
$\tau \leftarrow \tau + \delta$
$\mathcal{A}_t^\tau \leftarrow \mathcal{A}_t^{\tau+\delta}$
Low-level Motion Planner $P(a_{t+1})$
Local Path Planning
Pan-Tilt Control
Closed loop